When Modal Contact Rows Fail in Fixed-Budget XPBD: Diagnosis and an Operating Guide

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Abstract

Rigid-body XPBD engines already resolve contacts and joints, and a small global modal state (16–24 amplitudes) is an attractive way to give a stiff prop low-cost vibration without converting it into a full nodal or tetrahedral deformable. We show that inserting this globally shared modal state directly into a two-way unilateral contact row is not automatically energy-safe when the host runs only a few local iterations per substep. Across three scenes, two relaxations, and four schedules, our tested XPBD implementation overdraws the measured gross rigid-side supply by up to 4.4×10^7 J (a $2.2–4.5 \times 10^7$ J neighborhood over a deterministic perturbation ensemble), whereas augmented-Lagrangian and implicit implementations are markedly milder controls: at most 6.7 J in 3 of 24 cells, and never, respectively (an accounting audit places the control floor below 10^{-3} J, so the 6.7 J is real injection). An iteration ladder, equal-row-count schedules, warm starting, mode truncation, and a block-condensation ablation localize the amplification to finite-budget convergence of the shared modal coupling. When the contact architecture is flexible, the tested stiffness-aware implicit realization is the preferred observed control; in a fixed XPBD host the budget is best spent on iterations, with a verified basis, since more substeps and a block solve do not suffice. For hosts that still cannot converge the row, a cumulative modal-storage projection enforces its accounting bound unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells). Since the bound fixes only where the projected state lands, that projection can be chosen to hold the contact-observed surface fixed, cutting the penetration it opens from 2.4–47.6× the truncated solve's own to 1.0–6.3×: the residual is the price of the un-converged row, not of the guardrail, and this is containment rather than corrected contact.

CCS Concepts

• **Computing methodologies → Physical simulation**.

Keywords

contact, modal reduction, position-based dynamics, energy safety, fixed-budget solvers

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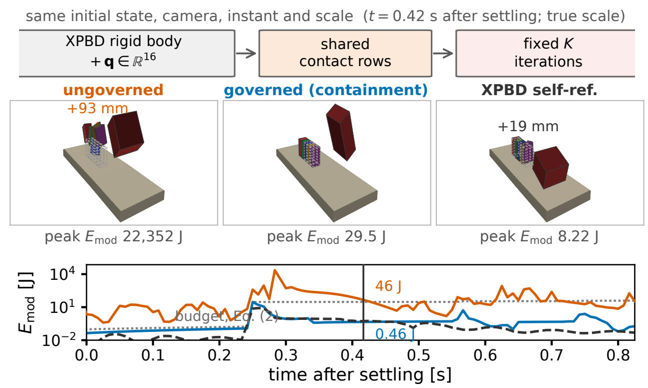

Figure 1: One shelf impact at a production-like interactive budget (1×8), true scale, three independent runs. Ungoverned, truncation energy throws the resting books up to 93 mm, more than their own height. Governed, (2) holds peak modal energy to 29.5 J and they stay within 1 mm of rest; the reference is the host's own high-iteration run (500×1). The bound removes the spurious launch and some legitimate motion with it: the reference lifts one book 19 mm, the governed run none.

1 Introduction

A developer building an interactive scene on a rigid-body XPBD engine [11] already has a solver that resolves contacts and joints under a small, fixed local-iteration budget. To make a stiff prop such as a shelf, a table, or a ledge ring when it is struck, one option is to replace it with a full nodal or tetrahedral deformable, at the cost of thousands of online coordinates and local internal constraints; the lighter alternative is to attach a small precomputed modal basis, 16–24 global amplitudes $\mathbf{q}$ that capture the low-frequency bending of a stiff, small-displacement body [8, 9, 13]. The runtime state grows by a handful of numbers, the precomputation is offline, and the same XPBD host steps everything. This paper is written for that developer; it does not argue XPBD against another solver.

The difficulty lies in the contact. In the real-time precedent the modal prop is driven *one way*: an external filter reads detected contact events and rings the modes with no back-reaction [3]. This double-counts the interface energy and cannot let the prop's own deformation change how it is supported. The physically consistent alternative is a *two-way* row, in which rigid and modal unknowns are resolved together through one shared contact multiplier [17], so that a heavier load bends the support and the bent support redistributes the load. The risk is that every support contact touches the same global vector $\mathbf{q}$: the contact-row graph is dense on the modes even though the state is small, and a host running only a few local iterations per substep need not converge that shared coupling. We take the row as given: at position level it is one linearization step from the velocity-level law ($C^{n+1} \approx C^n + h\,\mathbf{J}\mathbf{u}^{n+1}$, same Jacobian and multiplier as Zheng and James [17]), it is narrower than the original (normal rows only, e=0 restitution), and it is not a contribution of this paper. Our question is the practitioner's: what does the direct two-way transcription cost at a normal budget, and what should one do about it? Where solver architecture is unconstrained, our evidence favors the tested stiffness-aware velocity-level realization; our target, however, is the existing fixed-budget XPBD host.

Finite-iteration coupling across a partition boundary can inject spurious energy even when every subsystem is individually passive [15], and velocity-level frictional contact spanning rigid and reduced deformable bodies, with the energy artifacts of iterative solvers named, goes back at least to Kaufman et al. [10]. What has been missing is a quantitative account of the present case (unilateral contact-to-modal transfer in a fixed-budget host) and an operating guide for the practitioner who encounters it; this paper supplies both.

Contributions.

(1) **A practitioner workflow and its failure.** We treat adding a compact modal state to a fixed-budget XPBD host as the workflow it is, and show that the direct two-way contact row overdraws its measured rigid-side supply catastrophically at interactive budgets, across three scenes, two relaxations, and four schedules (§3.1), with augmented-Lagrangian and implicit implementations as milder controls. The cross-control reading is supported by a gravity/no-contact accounting audit, and the robustness of the failure by a deterministic perturbation ensemble.
(2) **Mechanism and an operating guide.** A same-path iteration ladder, equal-row-count schedules, warm starting, mode truncation, and an in-host block-condensation ablation localize the amplification to finite-budget convergence of the shared modal coupling (§3.2), and yield an ordered decision rule: the tested implicit realization when the architecture is flexible; otherwise, in a fixed XPBD host, iterations and a verified basis rather than more substeps or a block solve.
(3) **A last-resort guardrail and its price.** A cumulative modal-storage bound funded by the measured rigid-side loss and enforced by a scalar projection of the realized state (§4). It is a deliberately minimal fail-safe rather than a contact method: guaranteed unconditionally (Prop. 4.1, roundoff floor in all 90 measured cells), but at a measured cost to contact validity (§4.1).

Related work. Rigid XPBD [11], interactive modal deformation and velocity-level modal contact [8–10, 17], and reduced XPBD [13] set the stage; the last uses modes as a deformation surrogate, whereas here modal amplitudes enter as contact DOFs driven by the shared multiplier (cf. ray-traced penalty contact without a shared unilateral multiplier [18]). Our last-resort guardrail adapts energy-tank control [5, 7, 14] rather than inventing a new mechanism: we keep the tank-and-observer structure, but we fund the tank from the measured rigid-side loss rather than from the port it regulates, and we scale the realized state rather than a force, because a fixed-budget solver commits its multipliers before the excess is observable. These choices separate it from total-energy projection [4], bidirectional total-energy tracking [16], and the structural finite-budget passivity of bilateral couplings [15].

2 The row and the invariant it must respect

Let $\mathbf{q} \in \mathbb{R}^r$ be mass-normalized modal amplitudes of the supporting body, so its modal mechanical energy is $E_{\text{mod}} = \frac{1}{2}\dot{\mathbf{q}}^\top\dot{\mathbf{q}} + \frac{1}{2}\mathbf{q}^\top\mathbf{K}\mathbf{q}$. A support-contact row between a rigid corner at world height y_c and the deformed surface has gap

$$C \;=\; y_c \;-\; \big(y_{\text{rest}} + \mathbf{U}_y^\top\mathbf{q}\big), \qquad C \ge 0,\ \lambda \ge 0,\ C\lambda = 0, \qquad (1)$$

with $\partial C/\partial\mathbf{q} = -\mathbf{U}_y$, so one multiplier λ drives both the rigid body and the modal amplitudes. Equation (1) is the transcription discussed above, not a new law. It is a single row law, instantiated at every support contact in every substep (48 such rows on the shelf, 40 on the ledge, 200 on the table), with all rows sharing the one modal state $\mathbf{q}$.

The invariant. We require that the energy stored in the modal subsystem never exceed the measured gross rigid-side loss, defined exactly at (3) below, cumulatively over the run:

$$E^n_{\text{mod}} - E^0_{\text{mod}} \quad\le\quad \eta \sum_{k\le n} \max\!\big(\Delta E^k_{\text{rig}},\, 0\big), \qquad \eta \in (0,1]. \qquad (2)$$

The transfer efficiency η sets how much of the measured loss may be spent; every experiment in this paper runs $\eta = 1$, the least restrictive admissible value, so no result below depends on tuning it. The bound is cumulative rather than per-step: a substep may deposit more than it earns provided the running total remains within the budget, which is what allows a legitimate ring to persist across substeps.

Definitions. Modal storage is the mechanical energy of the reduced state as written above, and $E^0_{\text{mod}} = 0$: every scene starts from an undeformed, resting support, so even the run's own settling transient must be funded by the reservoir. The supply is

$$\Delta E^k_{\text{rig}} \;=\; \big(E^{k,-}_{\text{rig}} - E^{k,+}_{\text{rig}}\big) \;+\; W_g{}^k, \qquad W_g{}^k \;=\; \sum_b m_b\, \mathbf{g}\big(\mathbf{x}^{k,+}_b - \mathbf{x}^{k,-}_b\big), \quad (3)$$

where $E_{\text{rig}} = \sum_b \left[\frac{1}{2}m_b\|\mathbf{v}_b\|^2 + \frac{1}{2}\boldsymbol{\omega}_b^\top\mathbf{I}_b\boldsymbol{\omega}_b\right]$ is purely kinetic (over all dynamic bodies, not merely those in contact) and $\pm$ denote the two ends of substep k. We call this the **measured gross rigid-side loss**. It is a loss, not a dissipation: it also counts legitimate rigid-to-modal transfer and losses at rigid–rigid contacts elsewhere in the scene, so it is an envelope on what contact dissipated rather than a measurement of it (§6). Gravity enters only through W_g, so free ballistic motion registers zero supply, and only an inelastic velocity change funds the reservoir. Substep-boundary endpoints tile the timeline exactly, so no rigid energy change escapes the accounting.

What the invariant is, and two diagnostics. The bound (2) is a cumulative storage ceiling whose supply is measured rather than prescribed: the right-hand side is the measured loss of (3), never a tuned constant or a fraction of incident energy. It constrains a scalar reservoir, not a contact port, so it is neither port passivity nor a signed per-interface bound. The gross sum ignores substeps in which the rigid subsystem gains energy, and counts energy returned and re-dissipated as fresh supply (§6 measures both). We keep two diagnostics distinct: the incident-energy ratio R = peak modal energy / peak incident rigid KE, a severity measure, and the signed margin of (2) in joules, the invariant proper; they need not flag the same cells. Enforcing the bound requires a projection (§4).

3 The failure at a fixed budget

All solver-behaviour measurements below come from a single machine (Apple M4, CPU, CPython 3.12); every number is frozen with its generating command and commit hash in the supplemental ledger. We deliberately avoid mixing machines, because chaotic contact stacks diverge across architectures under floating-point reassociation.

3.1 One row, three implementations, the same 24 cells

We drive an impactor into a modally reduced support in three scenes (a shelf board, a ledge slab, and a 2.2×1.1 m table, the last two following the layouts of Coevoet et al. [3]) and sweep modal relaxation and the local budget (iterations × substeps) over {4×1, 8×2, 16×4, 32×8}. The hosts are XPBD [11], AVBD [6] (a descendant of vertex block descent [2]), and a sequential-impulse realization in the sense of Catto [1], made implicit in the modal block. All three are pinned to h=1/120 s, per-substep contact regeneration, modal rank 16/16/24 (shelf/ledge/table), Rayleigh damping, an implicit-midpoint modal restoring step, and η=1. They differ in the unknown solved for (position, position + multiplier, or velocity), in the modal contact weight (XPBD's explicit per-mode compliance $1/(H_{ii}h^2{-}1)$ versus the impulse host's implicit $(\mathbf{M}{+}h\mathbf{D}{+}h^2\mathbf{K})^{-1}$), in warm start, and in relaxation (the full host/parameter table is in the supplement). Two differences matter. Relaxation is inert on the impulse host (its implicit modal weight needs no under-relaxation), so its two relaxation rows are bit-identical. Warm start also differs ($\lambda \leftarrow 0$ every substep for the position-based host, carried duals for the others), so the comparison is as-deployed rather than compliance-matched. Warm start is not the lever, however: warm-starting the position-based host leaves the footprint unchanged (the same 8/24 cells, worst 1.20×10^5) and is worse at 4×1 (3.1×), since a warm multiplier cannot converge the shared modal DOF that only iterations converge (§3.2).

Equal row evaluations, unequal outcome. A cell costs $K \cdot S$ row evaluations per frame, so the budget axis spans ×64 and conflates two refinements that are neither interchangeable nor equally costly (a substep also repeats contact generation and integration). At 32 row evaluations on the shelf drop, spent as iterations (K=32, S=1) the ratio is 0.300 and (2) holds; spent as substeps (K=4, S=8) the ratio is 3.13, overdrawing by 481 J. Substep refinement alone does not converge the row.

Figure 2 shows the outcome. The position-based host exceeds $R = 1$ in 8 of 24 cells, worst case 1.19534×10^5 (ledge, relaxation 1.0, 4×1: peak modal energy 4.44×10^7 J against an incident 371 J); the augmented-Lagrangian host does so in 2 of 24, both at the starved 4×1 corner of the table scene with worst case 1.70, so it is not unconditionally benign either. The ordering is scene-dependent and can invert: taking the worst over scenes at 4×1, the position-based host is five orders of magnitude more energetic in the ratio R (1.2×10^5 vs 1.70), yet on the table scene itself the augmented-Lagrangian host is the worse (1.18 and 1.70 against a position-based worst of 0.37). Neither host dominates across scenes.

The ratio is not the invariant. We also measure (2) directly on all three hosts, ungoverned, running the reservoir accounting live while forcing $\gamma \equiv 1$: every state write in the enforcement path is guarded by $\gamma < 1$, so the trajectory is bit-identical to an ungoverned run (the ratios above reproduce the independent sweep exactly). The two diagnostics do not flag the same cells: the position-based host violates (2) in 9 of 24 against $R > 1$ in 8, the augmented-Lagrangian host in 3 against 2, and the implicit host in neither (0/24, worst margin at the roundoff floor), so neither diagnostic subsumes the other. What orders the failures is magnitude: the position-based overdraw reaches 4.4×10^7 J, seven orders beyond the largest augmented-Lagrangian one (6.7 J, at the starved 4×1 table corner) and beyond the implicit host's exact zero. A gravity/no-contact accounting audit puts that control floor below 10^{-3} J on both non-injecting hosts, so the augmented-Lagrangian overdraft is real impact-driven injection rather than accounting noise: three regimes that the ratio diagnostic compresses into one number.

3.2 The position-based amplification is a truncation artifact

Figure 3 isolates the local iteration budget. Two references recur below and are not the same object: the *implicit reference* is the implicit realization run to K=500 (the same code path, only more iterations), and the *XPBD self-reference* is the position-based host run to K=500 against itself. The position-based host falls monotonically from 2.96×10^4 at K=1 to 0.300 at K=32, approaching the implicit reference 0.2735; the gap |XPBD − reference| shrinks from 2.96×10^4 to 2.7×10^{-2}. The implicit realization moves only in the fourth decimal across $K = 2 \ldots 500$ (max−min 6.8×10^{-4}).

A falling energy ratio does not by itself show that the row is converging, since a single scalar could shrink coincidentally. Measuring the constraint directly resolves the question, and the two complementarity conditions must be read in their own units rather than compounded into one dimensionally mixed scalar. End-of-substep penetration falls from 27.9 mm at K=1 to 3.6 μm by K=64, where it saturates. The multiplier surviving on separated rows (force where the gap is open, in units of the steady-state median) instead rises from 84× to 139× through K=24 and collapses to exactly zero only at K=64. The energy crossing at $K \approx 24$ therefore coincides with the gap converging, not with complementarity holding; the multiplier side clears a rung later. These measurements concern the position-based host only: the augmented-Lagrangian multiplier is not the same object, and the implicit realization is converged at K=2.

Self-convergence stops short of the reference. Running the position-based host to K=500 against itself does not close the gap: its ratio plateaus by $K \approx 64$ at 0.2996, so the residual 2.6×10^{-2} against the reference's 0.2735 is a fixed point, not a tail. The state agrees no better: on the deflection trajectory $d_i(t) = \mathbf{U}_y[i]^\top \mathbf{q}(t)$ the peak matches to 2.4%, but the trajectories differ by 6.8 mm, 34% of the reference's own peak. The two hosts converge to different discrete solutions, as expected when they discretize the same continuous law differently. The sweep thus separates a truncation pathology that convergence removes from a discretization difference that it does not.

Two consequences follow. Convergence largely cures the amplification (iterating further or adopting the implicit realization removes

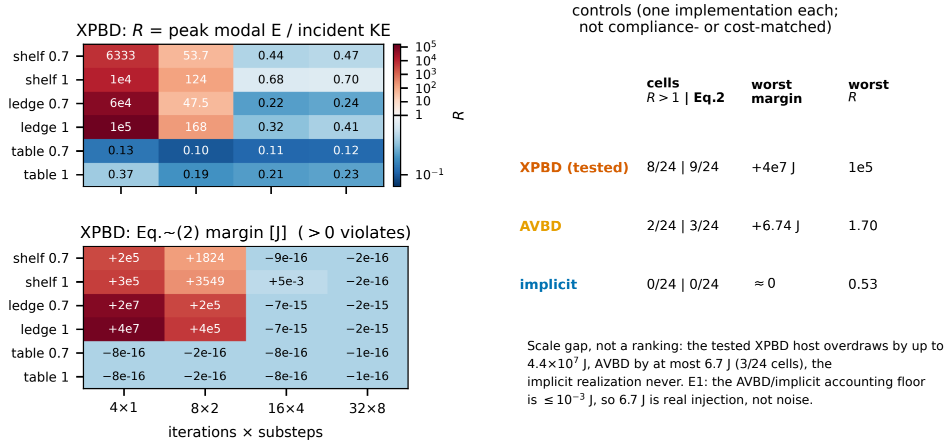

Figure 2: Our tested XPBD implementation, over the 24-cell sweep (3 scenes × relaxation $\{0.7, 1.0\}$ × budget), governor OFF. Left, top: R = peak modal energy / peak incident rigid KE, a severity diagnostic; bottom: the signed margin of (2) in joules (> 0 violates). The two need not flag the same cells (XPBD: R>1 in $8/24$, margin> 0 in $9/24$); the disagreement is a result, not noise. Right: a compact control strip for the three implementations (one of each, neither compliance- nor cost-matched): violating cells, worst margin, worst R. The comparison is a scale gap rather than a ranking: 4.4×10^7 J vs. ≤ 6.7 J vs. 0. The accounting audit (supplement E1) puts the control floor below 10^{-3} J, so 6.7 J is real injection. The full two-metric 72-cell three-implementation heatmap is in the supplement.

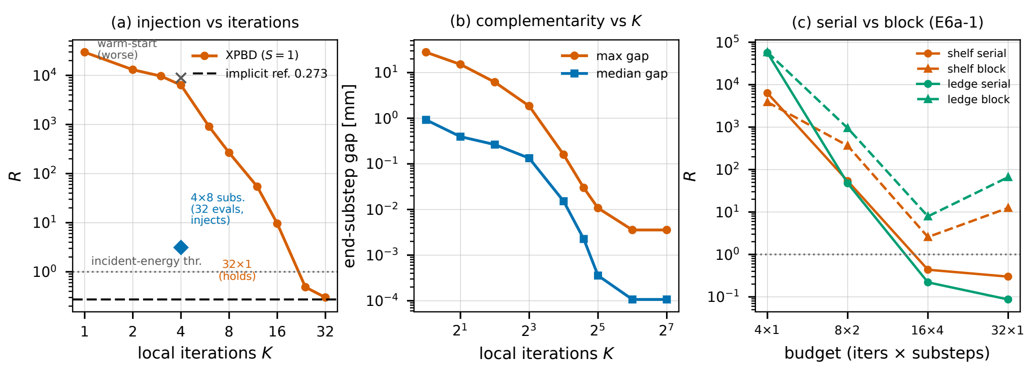

Figure 3: The XPBD operating envelope (our tested position-based host, governor OFF). (a) Injection R vs. local iterations K on a deterministic shelf drop (S=1): R decays monotonically toward the implicit reference (0.273), crossing the incident-energy threshold between K=16 and 24. Annotations mark the 32×1-vs-4×8 equal-row pair (the same 32 row evaluations still inject when spent as substeps) and warm starting (no improvement; worse at 4×1). (b) The end-of-substep gap converges with K, confirming that the falling ratio reflects the row converging rather than one scalar shrinking coincidentally. (c) Shared-row treatment (supplement E6a-1): serial per-row support projection (the path used in this paper) vs. joint block condensation, on both scenes. The serial path holds by 16×4–32×1; block condensation does not, following a different and worse convergence path that injects where the serial path holds. Condensing the shared rows is not the fix.

it), so the bound of §4 is not a substitute for convergence but targets the regime where neither escape is available. That regime is common: interactive position-based solvers run schedules near 1×8–2×4 (the one-iteration small-step regime [12]), well below the $K \geq 24$ this scene needs. On 18 deployed cells (3 scenes × 3 hosts, relaxation 0.7) the position-based host violates (2) in all six of its cells (worst R = 2282, on just 8 row evaluations), while the others match the sweep (1/6, 0/6).

Condensing the shared rows is not the fix. What converges the row is the iteration budget, not how the shared rows are scheduled: replacing the serial per-row support projection with a joint block condensation of the active rows (modal weight, restoring step, budget and warm start held fixed; Fig. 3(c)) does not remove the injection. It is comparable at the starved 4×1 corner but markedly worse at every higher budget where the serial path holds (R = 12.5 and 66.5 at 32×1 on shelf and ledge, against 0.30 and 0.09). Reproduced over two scenes and four budgets, it is a causal probe exposing a worse convergence path (a diagonal rigid-body block approximation), not a cure.

Robustness of the failure. The catastrophic cells are not a knife-edge. For each worst cell, the equal-row pair, and the guardrail cell we run 16 paired deterministic perturbations, 12 physical (drop height ±1–2%, normal speed, contact offset ±1.5 mm, tilt) and 4 contact-row-order permutations, kept as separate cohorts. Every injecting cell overdraws in all 16, the sign never flips, and the spread stays under one order of magnitude ($\log_{10}$ range 0.31–0.70); the 4.4×10^7 J worst case is the maximum of a 2.2–4.5×10^7 J band. The equal-row inversion and the scene dependence (never injecting on the table) survive too. Mode truncation localizes the amplification but does not remove it: excluding the stiff cluster cuts the shelf 4×1 ratio 6333 → 22.1 (584 J still overdrawn), and 6 of 8 injecting cells still overdraw (worst ledge $+1.24 \times 10^6$ J), so band-limiting the basis is not sufficient.

4 A last-resort guardrail

When a fixed host cannot converge the row and neither the implicit realization nor more iterations are available, a deliberately minimal fail-safe holds (2) regardless of the contact solve: it credits a reservoir from the measured supply and, when the realized modal state would overdraw it, projects that state back inside.

Enforcement. A reservoir B accumulates the supply and is debited by realized storage. Both happen after the contact solve, within the same substep: ΔE_{rig} is not observable until the velocity solve has finalized the rigid velocities, so the substep credits $B \leftarrow B + \eta \max(\Delta E_{\text{rig}}, 0)$ at that point, tests the realized modal energy against it, and only then debits $B \leftarrow \max(B - \max(\Delta E_{\text{mod}}, 0), 0)$. If the realized modal energy would overdraw B, the state is projected back into the admissible set $\{E_{\text{mod}} \le \bar{E}\}$, $\bar{E} = E^-_{\text{mod}} + B$, with $E^{\mp}_{\text{mod}}$ the modal energy before and after the solve.

Which admissible point is free, and the choice matters: scaling the whole state shrinks the load-bearing sag $\mathbf{U}_y^\top \mathbf{q}$ that resting bodies stand on, lifting the support into them (§4.1). We therefore project along the directions the active rows cannot see. Let $\mathbf{U}_c$ collect the support rows whose gap has closed, $\mathbf{d} = \mathbf{U}_c \mathbf{q}$, and $\mathbf{q}_{\text{qs}} = \mathbf{K}^{-1}\mathbf{U}_c^\top \mathbf{y}$ the least-elastic-energy state with the same $\mathbf{d}$, where $(\mathbf{U}_c \mathbf{K}^{-1} \mathbf{U}_c^\top)\mathbf{y} = \mathbf{d}$ and $E_{\text{qs}} = \frac{1}{2}\mathbf{d}^\top \mathbf{y}$. The remainder $\mathbf{q}_\perp = \mathbf{q} - \mathbf{q}_{\text{qs}}$ satisfies $\mathbf{U}_c \mathbf{q}_\perp = \mathbf{0}$ and $\mathbf{q}_{\text{qs}}^\top \mathbf{K} \mathbf{q}_\perp = 0$, so

$$(\mathbf{q}, \dot{\mathbf{q}}) \;\leftarrow\; (\mathbf{q}_{\text{qs}} + s\,\mathbf{q}_\perp,\;\; s\,\dot{\mathbf{q}}), \qquad s \;=\; \min\left(1,\; \sqrt{\frac{\bar{E} - E_{\text{qs}}}{E^+_{\text{mod}} - E_{\text{qs}}}}\right) \quad (4)$$

leaves $\mathbf{U}_c \mathbf{q}$ exactly unchanged while landing on $\bar{E}$. If $E_{\text{qs}} > \bar{E}$ the observed surface is itself unaffordable: the deepest-engaged prefix of rows that does fit is preserved instead (a bisection, E_{qs} being monotone in the row set), and failing that the state falls back to $(\beta \mathbf{q}_{\text{qs}}, \mathbf{0})$, $\beta = \sqrt{\bar{E}/E_{\text{qs}}}$, which is always feasible. With no active row, $\mathbf{q}_{\text{qs}} = \mathbf{0}$ and (4) is the whole-state scale $\gamma = \sqrt{\bar{E}/E^+_{\text{mod}}}$.

Each substep runs identically in all three hosts: snapshot $E^-_{\text{rig}}, E^-_{\text{mod}}$; predict and solve contact; form ΔE_{rig} ((3), post-solve) and credit $\eta \max(\Delta E_{\text{rig}}, 0)$ to B in the same substep; if $E^+_{\text{mod}} > \bar{E}$, project; debit B. Contact is not re-solved, and the active set is read from geometry alone, so no host-specific multiplier is needed. The projection is inert within budget (s=1, bit-identical trajectories).

Table 1: Where the penetration comes from: worst end-of-substep gap violation, position-based host, relaxation 0.7, measured identically in three arms. *unclamped* = governor off (the truncated solve's own penetration), *scaled* = whole-state projection, *preserved* = (4). Clamp count, corrective impulse (next substep's normal impulse over the unclamped steady-state median) and multiplier variance are the preserving arm's; 4×1 clamps nearly every substep, leaving no steady state (—).

			worst penetration [mm]				
scene	budget	clamped	unclamped	scaled	preserved	impulse	λ var.
shelf	4×1	104/108	6.16	**21.59**	8.31	—	—
shelf	8×2	201/216	0.50	6.43	0.50	6.9×	50×
ledge	4×1	90/108	8.80	**21.19**	17.79	—	—
ledge	8×2	166/216	0.91	4.08	0.93	8.8×	61×

Proposition 4.1. *For any $\eta \in (0, 1]$ and any host, the loop maintains $B \ge 0$ and guarantees (2) unconditionally, assuming nothing about the contact solve, up to an accounting tolerance $n\epsilon$ over n substeps (ϵ the per-test tolerance, 10^{-12} J).*

Proof sketch. Carry two invariants over substeps, with $c_k = \eta \max(\Delta E^k_{\text{rig}}, 0)$: $B_k \ge 0$ and $(E^k_{\text{mod}} - E^0_{\text{mod}}) + B_k \le \sum_{j \le k} c_j$ (both hold at k=0). Crediting c_k before the test caps realized storage at $E^-_{\text{mod}} + B_{k-1} + c_k$, so $\Delta E_{\text{mod}} \le B_{k-1} + c_k$; the debit keeps $B_k \ge 0$, and both invariants follow by induction. Since $B_n \ge 0$, this is (2) itself, not a relaxation: the same-substep credit absorbs the granularity term that a naive telescoping would leave. The argument uses only that the projected state lands in $\{E_{\text{mod}} \le \bar{E}\}$, never the direction it moves, which is what leaves (4) free to preserve the contact surface.

The governor holds. With the governor on, all 72 sweep executions (60 distinct configurations; the implicit host's relaxation pairs are bit-identical) and the 18 deployed cells of §3.2 satisfy (2): 90 governed cells at the roundoff floor (worst margin 3.6×10^{-15} J). Preserving the surface stores what the bound permits rather than the least it can, so the worst realized ratio is 1.15 (position-based) and 1.70 (augmented-Lagrangian), against 1.21 and 1.31 for a whole-state scale; R is a severity diagnostic, not the invariant, and the invariant holds in both. The projection is inert wherever the solve is already within budget: every implicit cell, and 16×4 and above on the position-based host at relaxation 0.7.

4.1 The cost of enforcement

Accuracy. Bounding the energy is useful only if the bounded solution improves on the unbounded one. At a moderate injecting cell (shelf, 8×2, relaxation 0.7) peak modal energy is 1555.6 J ungoverned and 28.7 J governed against the implicit reference's 7.92 J: the energy error falls from 196× to 3.6×, and 189× to 3.5× against the XPBD self-reference, so it does not depend on the reference. The trajectory still degrades, but far less than under a whole-state scale: $\|d - d^{\text{ref}}\|_\infty$ rises from 6.5 mm (33% of the reference peak) to 9.2 mm (46%), against 14.2 mm (71%) when scaling; the surviving sag is 12.5 mm against the reference's 20.5, where scaling leaves 6.0. The governor is still a safety envelope, not an accuracy device.

This is also why a 196× energy error coexists with a peak deflection only 1.2× too large: 99.6% of the excess sits in the shelf's stiff mode cluster above 20 kHz (~200× the substep rate). A whole-state scale cannot separate that from the legitimate sag and removes the amount but not the character of the excess (99.6 → 97.7% above the cut); projecting along the contact-invisible directions removes proportionally more (99.6 → 89.8%), the sag it protects being exactly the low-frequency content. Band-limiting the basis remains the first-line remedy.

Contact validity. Any projection that rewrites **q** after the solve moves the support surface without re-solving contact, so it can open a gap violation. Table 1 separates the contributions, instrumented by wrappers that leave the solve unperturbed (clamp counts reproduce the independent sweep exactly).

The penetration is not created by the guardrail; it is the price of the truncated row. At converged budgets no substep clamps and the worst violation is 0.011–0.124 mm (16×4, 32×1, both scenes), so the artifact exists only where the row is truncated; at starved budgets the unclamped solve already opens 0.17–8.8 mm with no governor present. What a projection does is *amplify* that residue: a whole-state scale by 2.4–47.6× over the eight cells of both relaxations, reaching 21.6 mm (72% of the 30 mm board); preserving the observed surface only 1.0–6.3×, a 1.2–12.9× improvement that lands within 3% of the unclamped solve in two of four relaxation-0.7 cells. The exception is ledge 4×1 (17.8 mm), where at the worst substeps the observed surface would cost over 300× the whole budget: that surface is itself the injection artifact, and no bound-respecting projection can keep it.

The corrective transient does not improve with it: the next substep's normal impulse is 6.9–8.8× the steady-state value and multiplier variance 50–61× higher in the 8×2 cells, against 3.7–8.9× and 5.6–62× when scaling the whole state. Fewer millimetres, not a gentler recovery. Rigid momentum is unchanged (exactly zero, verified), since (4) writes only modal state (full rows in the supplement).

5 Fidelity and runtime

5.1 The reduced response against a full-FEM reference

Modes are worth using only if the reduced response is faithful. Against an unreduced FEM solve on the same discrete operator (so discretization bias cancels), the reduced ledge response tracks the reference: the peak-deflection ratio rises 0.38 → 0.54 → 0.79 → 0.88 as h refines 1/120 → 1/960 (the residual ~10% is k=24 truncation, not coupling error), ring frequency agrees to 0.4% (78.0 vs 78.3 Hz), and the far-field falloff tracks at Spearman $\rho = 0.89$ with no fitted attenuation.

5.2 Runtime cost

On the CPU host the ledger adds 0.23–0.41 ms at 16×4, or 0.9–3.4% of an 11–126 ms baseline (the table scene's cost sits below its run-to-run variance, so we report the percentage); the split of (4) adds one $m \times m$ solve per clamped substep, $m \leq 8$ active rows here. The relative cost is larger where the budget is tight: at the deployed 1×8/2×4 schedules the same ~0.3–2.2 ms is 6.6–34.3%. We make no unqualified real-time claim; an enforced device-resident projection does not exist here (§6).

6 Limitations

Stability is not accuracy. Where the projection is load-bearing it still opens contact violations (up to 17.8 mm) and moves the trajectory away from the implicit reference even as it moves the energy toward it (§4.1): bounded, not faithful. Its proper use is a safety envelope for budgets that would otherwise diverge; convergence is the better remedy wherever affordable. Preserving the observed surface narrows that gap without closing it, and does not reduce the corrective transient at all. Selecting preserved rows by a load-weighted subspace rather than whole rows, and one-sided constraints, both target the regime where it fails; neither is built.

Scope of the row. The coupling is normal-only (friction stays body-local, so scraping and rolling transfer lie outside it), and the position-level form fixes e=0. The bound governs modal storage only, not energy created spuriously in rigid DOFs: in the impulse realization, box–box rows rectify a support ring into net rigid motion, to which the ledger is blind by design.

Supply and coverage. The supply is a gross sum, so energy returned to the rigid bodies and re-dissipated is credited twice. We bound the exposure by the opposite channel, $\sum_k \max(-\Delta E^k_{\mathrm{rig}}, 0)$ as a fraction of supply: 3–32% (position-based), 0.4–27% (implicit), and 102–118% on the augmented-Lagrangian host (whose rigid subsystem gains more than it loses in every cell, the box–box effect above). All of these are above the ≲1% expected and partition-dependent, so the budget belongs to the schedule as well as the scene. Enforcement is host-side and single-machine (CPU float64); a device-resident governor and a governed-path FEM validation remain future work. We test three formulations with one implementation of each, and three scenes in one contact regime; the augmented-Lagrangian host still overdraws in 3/24 cells, so any claim that descent-class solvers are passive here is empirical about this sweep, not a theorem.

7 Conclusion

For a developer on a fixed-budget XPBD host wanting low-cost vibration on a stiff prop, the decisions are ordered; the question is not XPBD versus another solver. First pick the representation: full-space XPBD for large, local or nonlinear deformation, a compact modal state for stiff small-displacement ringing. If the contact architecture is still open, the tested stiffness-aware implicit realization is the cleanest control we observed (it never overdraws), an implementation-specific observation, not a solver-class theorem. If the XPBD host is fixed, the direct two-way modal row overdraws its measured supply by up to 4.4×10^7 J at a few local iterations, and neither more substeps at equal row cost, nor warm starting, nor block-condensing the shared rows removes it (§3.2); iterations do, so spend the budget there and verify the basis. Only when a fixed host still cannot converge does the cumulative modal-storage bound earn its place: it holds in all 90 measured cells, and projecting so as to preserve the contact-observed surface keeps the penetration it adds to 1.0–6.3× what the truncated solve opens unaided. Containment, not corrected contact: a modal extension is cheap in state size, but not automatically easy for a truncated solve.

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